Prevalence and severity of sleep apnea in a group of morbidly obese patients.

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Obesity is the most important risk factor for obstructive sleep apnea. It is estimated that 70% of sleep apnea patients are obese. In the morbidly obese, the prevalence may reach 80% in men and 50% in women. The aim of this study was to determine the prevalence and severity of sleep apnea in a group of morbidly obese patients, leading to bariatric surgery. In a cross-sectional study developed in Bahia, northeastern Brazil. 108 patients (78 women and 30 men) from the Obesity Treatment and Surgery Center--"Núcleo de Tratamento e Cirurgia da Obesidade" underwent standard polysomnography. Patients with an apnea-hypopnea index (AHI) > or = 5 events/hour were considered apneic. Mean +/- SD for age and BMI were 37.1 +/- 10.2 years and 45.2 +/- 5.4 kg/m2, respectively. The calculated AHI ranged widely from 2.5 to 128.9 events/hour. Sleep apnea was detected in 93.6% of the sample, wherein 35.2% had mild, 30.6% moderate and 27.8% severe apnea. Oxyhemoglobin desaturation was directly related to the AHI and was more severe in men. There was a high frequency of sleep apnea in this group of morbidly obese patients, for whom it was very important to request polysomnography, thus enabling therapeutic management and prognostication.

The relationship of obesity and obstructive sleep apnea.

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Obstructive sleep apnea is a common disorder, and obesity is a known risk factor for its development. The prevalence of obesity is increasing worldwide, and a corresponding increase in the prevalence of obstructive sleep apnea and its cardiovascular and noncardiovascular consequences is likely. This article reviews the established evidence supporting obesity as a risk factor for obstructive sleep apnea and discusses the evidence suggesting that obesity is also a consequence of obstructive sleep apnea. There is evidence that treating obesity reduces the severity of obstructive sleep apnea and that treating obstructive sleep apnea decreases obesity. However, the evidence does not support a sustained correlation between weight loss and improvement in sleep-disordered breathing.

Obesity, sleep apnea syndrome, and rhythmogenic risk.

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Obstructive sleep apnea is a common disorder and affects approximately 4% of middle-aged men and 2% of middle-aged women. Obstructive sleep apnea is clearly associated with obesity, with more than 50% of patients having a body mass index>30 kg/m2. Substantial evidence identified obstructive sleep apnea as risk factor not only for excessive daytime sleepiness and road traffic accidents, but also for increased cardiovascular morbidity and mortality. In addition, all kinds of arrhythmias have been observed in patients with sleep apnea ranging from asymptomatic sinus bradycardia to sudden cardiac death. Approximately 5-10% of patients with obstructive sleep apnea show marked apnea-related bradyarrhythmias due to enhanced vagal tone and pronounced hypoxia. Therapeutic options in obese patients with obstructive sleep apnea include consequent weight loss and nasal continuous positive airway pressure (CPAP) ventilation as the therapy of first choice. Weight reduction and effective nasal CPAP therapy significantly decrease cardiovascular morbidity and mortality and eliminate sleep-related

bradyarrhythmias in 80-90% of patients obviating the need for pacemaker implantation in these patients.

Shared genetic risk factors for obstructive sleep apnea and obesity.

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Both obesity and obstructive sleep apnea (OSA) are complex disorders with multiple risk factors, which interact in a complicated fashion to determine the overall phenotype. In addition to environmental risk factors, each disorder has a strong genetic basis that is likely due to the summation of small to moderate effects from a large number of genetic loci. Obesity is a strong risk factor for sleep apnea, and there are some data to suggest sleep apnea may influence obesity. It is therefore not surprising that many susceptibility genes for obesity and OSA should be shared. Current research suggests that approximately half of the genetic variance in the apnea hypopnea index is shared with obesity phenotypes. Genetic polymorphisms that increase weight will also be risk factors for apnea. In addition, given the interrelated pathways regulating both weight and other intermediate phenotypes for sleep apnea such as ventilatory control, upper airway muscle function, and sleep characteristics, it is likely that there are genes with pleiotropic effects independently impacting obesity and OSA traits. Other genetic loci likely interact with obesity to influence development of OSA in a gene-by-environment type of effect. Conversely, environmental stressors such as intermittent hypoxia and sleep fragmentation produced by OSA may interact with obesity susceptibility genes to modulate the importance that these loci have on defining obesity-related traits.

Obesity and hormonal factors in sleep and sleep apnea.

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The presence of obesity, defined as weight 20 per cent or more above ideal body weight or increased body fat content, significantly increases risk of pulmonary, cardiovascular, metabolic, and gastrointestinal problems. Obesity is a major cause of shortened life expectancy. While obesity is not essential for the development of the obstructive sleep apnea syndrome, a significant percentage of patients with obstructive sleep apnea are obese. When evaluating these patients who have obstructive sleep apnea, it is important to search diligently for medical problems that are commonly found among the obese. While there is an increased incidence of obese patients among those who have obstructive sleep apnea, the exact reason for this is uncertain. The study of endorphins and enkephalins may expand our understanding of obesity, ventilatory regulation, and obstructive sleep apnea. This may, in fact, enable us to understand better the interrelationship between obesity and obstructive sleep apnea. The role that thyroid hormone, testosterone, and progesterone play in obstructive sleep apnea has also been reviewed. Patients who have obstructive sleep apnea should not be treated with testosterone. All patients given testosterone should be observed quite closely for the possible signs and symptoms of obstructive sleep apnea. Progesterone seems to be of some help in patients who have obesity hypoventilation syndrome. Its effectiveness in patients with obstructive sleep apnea is less clear. The obesity hypoventilation syndrome as described by Burwell is relatively uncommon. Many of the manifestations of the obesity hypoventilation syndrome, however, are found in patients with obstructive sleep apnea. The recognition that the symptoms stem from underlying obstructive sleep apnea offers great potential for therapy. Weight reduction is valuable therapy for patients with obesity and pulmonary dysfunction, obesity and obstructive sleep apnea, and obesity hypoventilation syndrome. Weight reduction and weight maintenance, while difficult, are essential in patients with obesity, obesity and obstructive sleep apnea, and the hypoventilation syndrome. Obesity should be viewed as a medical problem deserving medical attention and long-term medical follow-up.

[Sleep apnea syndrome and obesity hypoventilation syndrome].

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Obstructive sleep apnea (OSA) is characterized by repeated episodes of apnea and hypopnea during sleep. Obesity is the most important risk factor for OSA. From the recent reports, the prevalence of OSA is high. Data have shown that severe OSA (the mean number of apnea and hypopnea episodes per hour slept: AHI > or = 30) is a risk factor for cardiovascular diseases and death. In addition, OSA is widely prevalent in patients with obesity, diabetes, and hypertension which are the major risk factors for cardiovascular disease. Several reports show that sleep duration has significant effects on BMI, mortality and diabetes. It has been reported that leptin and ghrelin, which were the two key hormones in appetite, are elevated in OSA patients. Therefore, it is important to investigate the relationship between OSA, body weight gain and obesity.

Risk of Sleep Apnea Is Associated with Abdominal Obesity Among Asian Americans: Comparing Waist-to-Hip Ratio and Body Mass Index.

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This study examines associations between the risk of sleep apnea and abdominal obesity (assessed by waist-to-hip ratio (WHR)) and general obesity (assessed by body mass index (BMI)) in a sample of Chinese and Korean American immigrants. The dataset included Chinese and Korean participants aged 50-75 who were recruited from primary care physicians' clinics from April 2018 to June 2020 in the Baltimore-Washington D.C. Metropolitan area ($n = 394$). Abdominal obesity was determined if $WHR \geq 0.9$ in men and $WHR \geq 0.85$ in women. General obesity was determined if $BMI \geq 30$. The risk of sleep apnea was determined by using the Berlin questionnaire. Poisson regression models examined associations between sleep apnea risk and obesity. Models controlled for socio-demographic risk factors. Twelve percent of the study participants were classified as a high risk for sleep apnea, and 75% had abdominal obesity whereas 6.4% had general obesity. High risk of sleep apnea was positively associated with abdominal obesity ($PR = 1.31$, 95% CI: 1.17-1.47) and general obesity ($PR = 2.19$, 95% CI: 0.90-5.32), marginally significant at $p < 0.1$). Chinese and Korean immigrants living in the USA who are at high risk of sleep apnea have higher abdominal obesity, even after accounting for sociodemographic characteristics. Abdominal obesity may be a better indicator than general obesity when examining the risk of sleep apnea among Asian Americans. Name: Screening To Prevent ColoRectal Cancer (STOP CRC) among At-Risk Asian American Primary Care Patients NCT Number: NCT03481296; Date of registration: March 29, 2018 URL: https://clinicaltrials.gov/ct2/show/NCT03481296?term=Sunmin+Lee&draw=2&rank=1.

Causal Relationship Between Childhood Obesity and Sleep Apnea Syndrome: Bidirectional Two-Sample Mendelian Randomization Analysis.

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Childhood obesity has become a global pandemic, leading to a range of diseases. Childhood obesity appears to be associated with an increased prevalence of sleep apnea syndrome. Sleep apnea is an inestimable risk factor for thrombosis, hypertension, cardiomyopathy and many other diseases. Therefore, exploring the relationship between childhood obesity and sleep apnea syndrome will help to understand the potential link between the two and provide research directions for future disease prevention and treatment. However, no studies have confirmed whether there is a causal relationship between childhood obesity and sleep apnea syndrome. The IEU OpenGWAS project provided the GWAS-aggregated data for childhood obesity and sleep apnea syndrome. Inverse-variance weighted (IVW) was used as the main method to evaluate the causal relationship between childhood obesity and sleep apnea syndrome. Single nucleotide polymorphisms (SNPs) were regarded as instrumental variables, and the screening threshold was $P < 5.0\times10^{-6}$. Leave-one-out method was performed to confirm the robustness of the results. IVW analysis confirmed a causal relationship between genetic susceptibility to childhood obesity and an increased risk of sleep apnea syndrome [odds ratio (OR)=1.12, 95% confidence interval (CI): 1.02-1.23, P=0.016]. However, two-sample MR results also showed no causal relationship between genetic susceptibility to sleep apnea syndrome and an increased risk of childhood obesity (OR=1.50, 95% CI: 0.95-2.38, P=0.083). The intercept of MR-Egger regression was close to 0, which implies that there are no confounding factors in the analysis to affect the results of two-sample MR analysis. The leave-one-out results show that the bidirectional two-sample MR analysis results were robust. There is a causal relationship between genetic susceptibility to childhood obesity and increased risk of sleep apnea syndrome. People with a history of childhood obesity should pay more attention to physical examination to early prevention and management of sleep apnea syndrome.

Global burden of sleep-disordered breathing and its implications.

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One-seventh of the world's adult population, or approximately one billion people, are estimated to have OSA. Over the past four decades, obesity, the main risk factor for OSA, has risen in striking proportion worldwide. In the past 5 years, the WHO estimates global obesity to affect almost two billion adults. A second major risk factor for OSA is advanced age. As the prevalence of the ageing population and obesity increases, the vulnerability towards having OSA increases. In addition to these traditional OSA risk factors, studies of the global population reveal select contributing features and phenotypes, including extreme phenotypes and symptom clusters that deserve further examination. Untreated OSA is associated with significant comorbidities and mortality. These represent a tremendous threat to the individual and global health. Beyond the personal toll, the economic costs of OSA are far-reaching, affecting the individual, family and society directly and indirectly, in terms of productivity and public safety. A better understanding of the pathophysiology, individual and ethnic similarities and differences is needed to better facilitate management of this chronic disease. In some countries, measures of the OSA disease burden are sparse. As the global burden of OSA and its associated comorbidities are projected to further increase, the infrastructure to diagnose and manage OSA will need to adapt. The use of novel approaches (electronic health records and artificial intelligence) to stratify risk, diagnose and affect treatment are necessary. Together, a unified multi-disciplinary, multi-organizational, global approach will be needed to manage this disease.

Childhood and Adolescent Obesity: A Review.

PMID: 33511092 | Indexed in PubMed

Obesity is a complex condition that interweaves biological, developmental, environmental, behavioral, and genetic factors; it is a significant public health problem. The most common cause of obesity throughout childhood and adolescence is an inequity in energy balance; that is, excess caloric intake without appropriate caloric expenditure. Adiposity rebound (AR) in early childhood is a risk factor for obesity in adolescence and adulthood. The increasing prevalence of childhood and adolescent obesity is associated with a rise in comorbidities previously identified in the adult population, such as Type 2 Diabetes Mellitus, Hypertension, Non-alcoholic Fatty Liver disease (NAFLD), Obstructive Sleep Apnea (OSA), and Dyslipidemia. Due to the lack of a single treatment option to address obesity, clinicians have generally relied on counseling dietary changes and exercise. Due to psychosocial issues that may accompany adolescence regarding body habitus, this approach can have negative results. Teens can develop unhealthy eating habits that result in Bulimia Nervosa (BN), Binge- Eating Disorder (BED), or Night eating syndrome (NES). Others can develop Anorexia Nervosa (AN) as they attempt to restrict their diet and overshoot their goal of "being healthy." To date, lifestyle interventions have shown only modest effects on weight loss. Emerging findings from basic science as well as interventional drug trials utilizing GLP-1 agonists have demonstrated success in effective weight loss in obese adults, adolescents, and pediatric patients. However, there is limited data on the efficacy and safety of other weight-loss medications in children and adolescents. Nearly 6% of adolescents in the United States are severely obese and bariatric surgery as a treatment consideration will be discussed. In summary, this paper will overview the pathophysiology, clinical, and psychological implications, and treatment options available for obese pediatric and adolescent patients.